

Pranav Kunnamkudath Roy

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EDUCATION

SRM Institute of Science and Technology <i>B.Tech., CSE – Computer Networking</i> CGPA: 8.33/10	Kattankulathur, Tamil Nadu 2023 – 2027
Mount St. Mary's School <i>Class XII, CBSE</i> Percentage: 75.4%	New Delhi 2023
Mount St. Mary's School <i>Class X, CBSE</i> Percentage: 91.4%	New Delhi 2021

PROJECTS

RWM – Rust Window Manager <i>Rust, x11rb, X11</i> github.com/ReitardXd/rwm - Built a minimal tiling WM from scratch in Rust using raw X11 protocol bindings; no external WM frameworks. - Master-stack tiling, focus-follows-mouse via EnterNotify, stack rotation, and keybinding dispatch with modifier masking. - Graceful window kill via WM_DELETE_WINDOW protocol with XKillClient fallback; full ConfigureRequest passthrough.	2025
NERVE – Network Fault Detection <i>Python, Cisco DNAC API, Groq LLaMA, Streamlit, SQLite</i> - Automated network incident management system polling Cisco DNAC for real-time fault telemetry with remediation workflows. - Integrated Groq LLaMA for AI-driven root cause analysis; Discord webhook alerts and Streamlit dashboard for live visualization.	2025
N-Body Gravitational Simulator <i>C++, OpenGL, GLSL, GLFW</i> github.com/ReitardXd/nbody 3D N-body simulator with modern OpenGL pipeline: bloom post-processing, particle trails, orbital camera; no physics libraries. - Spiral galaxy procedural initial conditions; rogue black hole mode; Octree stub for Barnes-Hut $O(n \log n)$ approximation.	2024
ProctorAI <i>FastAPI, PostgreSQL, Docker, Python, Computer Vision</i> github.com/ReitardXd/ProctorAI - AI-based remote proctoring platform with FastAPI backend, PostgreSQL persistence, and Docker containerization. - Real-time gaze tracking and behavioral anomaly flagging for exam integrity monitoring during live sessions.	2024
Rei-Shell – POSIX Shell in C <i>C, POSIX</i> github.com/ReitardXd/Rei-Shell POSIX-compliant shell with process forking, exec, piping, redirection, signal handling, and environment variable expansion.	2024
cracksmoker <i>Python, Tkinter, hashcat, multiprocessing</i> github.com/ReitardXd/cracksmoker hashcat frontend GUI with live output streaming, mask builder, and support for MD5/SHA1/SOA256/bcrypt/SOA-512 attack modes.	2024

TECHNICAL SKILLS

Languages: C++, C, Rust, Python, JavaScript, Bash/Shell, SQL
Systems & Networking: POSIX, Linux (Arch), TCP/IP, X11, Cisco DNAC API
Frameworks & Tools: FastAPI, Streamlit, OpenGL/GLFW, Node.js, PostgreSQL, Docker, Git, Neovim, GDB
Security: HackTheBox, CTF, hashcat, vulnerability research
Competitive Programming: Codeforces – Specialist, LeetCode

PUBLICATIONS

Smart Sleep Posture Monitoring System Using Embedded Sensors and IoT - International Research Journal on Advanced Engineering and Management - ESP32-based sleep posture classifier using MPU6050 IMU and FSR sensors with ML classification and vibration feedback.	May 2025
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LANGUAGES

English (Native) | Hindi (Native) | Tamil (Professional Working) | Malayalam (Professional Working)